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Rheological Measurement Dataset of Resin and Composite Mixtures for Digital Light Processing based Additive Manufacturing

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Abstract

This work features a dataset intended to aid in the development of processing parameters for particulate composite materials for additive manufacturing (AM) using digital light processing (DLP) method. Hollow glass microspheres (HGMs) are used as particulate fillers to manufacture syntactic foam composites. The standard thermosetting resins have a recommended set of 3D printing parameters. However, when the particles are mixed to 3D print composites, the mixture properties change, and the processing parameters need to be adjusted according to the particle volume fraction, size and other parameters. This dataset provides rheological measurements on the neat resin and resin–HGM composite mixtures intended for vat photopolymerization (VP) based DLP method. Three HGM density grades (0.13, 0.23, and 0.31 g/cm³ true particle densities) were examined at 10, 20, and 40 vol.% to capture the effects of particle density and volume fraction on resin flow behavior. Viscosity was measured using an Anton Paar MCR702 analyzer at a range of shear rates (1–50 s⁻¹) and temperatures (35–125 °C), generating 50 independent data files for 5 trials repeated for each of the 10 compositions. The dataset records the coupled influence of temperature and shear rate for each composition, providing a foundation for modeling viscosity–shear rate–temperature–composition relationships. Interpolations of various parameters at a benchmark viscosity of 0.40 Pa·s enable extraction of processing windows relevant to VP of the composite mixtures and significantly reduce the trial and error involved in processing parameter optimization for composite mixtures. The dataset also includes μ CT image stacks for representative specimens, supporting qualitative assessment of particle dispersion and internal structure.

Keywords: Additive manufacturing; Digital light processing; Syntactic foam; Rheological analysis

Background & Summary

Syntactic foams are used as structural and buoyancy materials in marine applications, especially in human or automated underwater vehicles [1]. These hollow-particle filled close-cell foams are strong under compression and have low moisture absorption, making them suitable for underwater applications [2]. The performance standards for such materials are continuously increasing as the vehicles are being designed for greater depths. Traditionally, cast syntactic foam blocks are machined and joined together to make the vehicle structures [3], but deeper missions require elimination of joints to reduce stress concentration locations and additive manufacturing (AM) has

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emerged as the promising method for such structures [4]. Compatibility of resin and particles, printing method, and scalability are among the parameters that need to be considered in designing new AM methods for such materials [5]. Filaments of thermoplastic resin syntactic foams are available for extrusion-based AM methods. However, development of additive manufacturing methods for thermosetting resin-based syntactic foams remains challenging due to particle-resin density mismatch, which can lead to particle segregation, as well as the complex rheological behavior of highly filled suspensions that complicates process parameter optimization. While slurry optimization is well established in other fields such as ceramics processing, the coupled effects of photopolymerization kinetics, temperature-dependent viscosity, and particle loading in vat photopolymerization systems remain less well understood [6-8].

Vat Photopolymerization (VP) has recently emerged as a promising method for developing thermosetting resin matrix syntactic foams [9, 10]. The UV responsive acrylate resin used in VP has a density of 1.1-1.2 g/cm³. Whereas the density of most hollow particles is well below 1 g/cm³ [11]. This density mismatch leads to the particle floating and creates a gradient in the finished piece [12]. It is well established that increasing particle content leads to increased viscosity in particulate suspension due to hydrodynamic interactions and particle crowding, while increasing temperature reduces viscosity through decreased intermolecular resistance and enhanced molecular mobility [13, 14].

Rheological analysis can determine the optimal temperature and mixing rate that can be used in VP to manufacture high quality syntactic foam by mapping these two quantities [15]. Because VP processing is sensitive to resin flow behavior, benchmark viscosity can serve as a practical reference point for comparing neat resin and particle-filled mixtures under processing relevant conditions. In this study, the room temperature viscosity of the neat resin, 0.40 Pa·s, was used as a reference value to identify combinations of temperature and shear rate that produce comparable flow behavior in the composite mixtures. This benchmark is used here as a practical processing metric rather than as a complete description of printability, which can also depend on factors such as cure kinetics, and material stability. The hollow particles used in manufacturing syntactic foams are usually in a narrow range of average particle size. Changing the volume fraction of particles should follow similar trends in viscosity regardless of the true particle density of the particles of the same diameter. Dynamic mechanical analysis, including rheological measurements, allows for deeper exploration of a material's properties by examining the shear rate- temperature dependencies of a material. These often output a lot of data, making them good candidates for machine learning [16, 17].

The present dataset captures viscosity as a function of temperature and shear rate for resin and a variety of composite mixtures. The temperature varied over 10 increments from room temperature to 120 °C. The shear rate varied in 25 increments from 1 to 50 s⁻¹. Each of the 10 material compositions were characterized 5 times, generating a total of 50 raw data files with temperature, shear rate, viscosity, rotational speed, and the material composition as variables. Viscosity values are recorded in centipoise (cP) from rheometer output and format of the provided dataset; the corresponding SI units and analysis was performed using Pascal seconds (Pa·s). The presented data can be useful for developing and validating theoretical models, training machine learning

algorithms, and 3D printing syntactic foams using the optimized process parameters. A related study by Beckwith et al (2025) investigated the application of rheology-guided processing for additive manufacturing of syntactic foams using a subset of the data presented here [18]. In addition to rheological measurements, μ CT imaging datasets are included to provide supporting microstructural information for the tested material systems. In contrast, the present work provides the complete rheological dataset, including raw measurements across multiple material systems, temperatures, and shear rates. This Data Descriptor is intended to make these datasets publicly accessible and reusable for a wide range of applications beyond the scope of the prior study.

Materials and Methods

Raw materials

Hollow glass microspheres (HGMs), commercially marketed as “Ecospheres” manufactured by Trelleborg Offshore, Boston, were used as particulate fillers in this study. Three types of HGMs with mean particle size of 45 μm and manufacturer-reported particle densities of 0.13, 0.23 and 0.31 g/cm^3 are used in the work. A syntactic foam microstructure containing HGMs of 0.31 g/cm^3 is shown in Figure 1. With the glass material density of about 2.5 g/cm^3 , glass modulus of 70 GPa and Poisson’s ratio of 0.2, the wall thickness, the radius ratios (defined by the ratio of the inner radius to the outer radius of the HGMs, $\frac{r_i}{r_o}$), and the effective modulus (E_{mb}) of the HGMs are calculated respectively and shown in Table 1. The radius ratio provides a measure of shell thickness relative to particle size and is commonly used to characterize hollow particle geometry. E_{mb} is calculated using equation obtained from [19]. The equation to calculate E_{mb} follows

$$E_{mb} = E_s \frac{1 - \left(\frac{r_i}{r_o}\right)^3}{1 + 2\left(\frac{r_i}{r_o}\right)^3} \quad (1)$$

Where E_s is the modulus of the glass shell material, and r_i and r_o are the inner and outer radii of the microspheres respectively. The outer radius was determined from the reported mean particle size, and the inner radius was calculated from the radius ratio. The glass material properties were obtained from literature values. These HGMs are weak under shear loading, which is usually the failure mode under uniaxial compression [20].

The neat resin used in this work is ABS-Like Translucent Resin produced by Elegoo, China. The data sheet provided by the manufacturer lists that the resin should be printed at room temperature, with a stated viscosity of 0.40 $\text{Pa}\cdot\text{s}$ at 25 $^\circ\text{C}$ [21]. Each steady shear rheological measurement used 2 mL of resin with HGM content of 10, 20, and 40 vol.%. The true particle densities of HGMs were 0.13, 0.23, and 0.31 g/cm^3 .

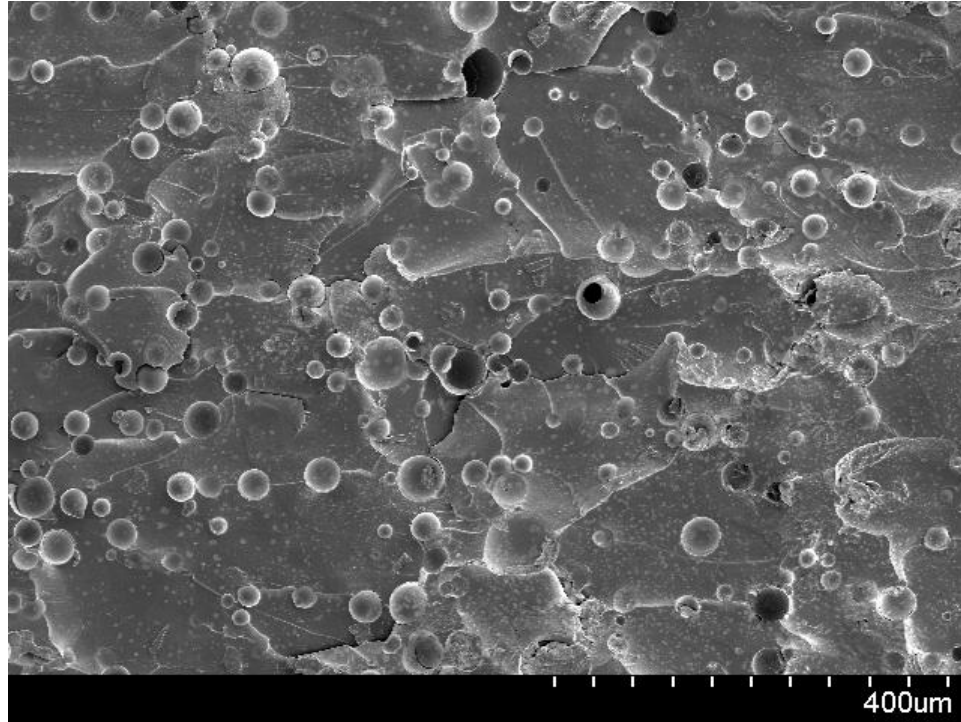


Figure 1. SEM image of a syntactic foam specimen filled with HGMs of 0.31 g/cm³ true particle density. The image is obtained from a freeze fractured surface.

Table 1. Properties of three types of HGMs.

True Particle Density (g/cm ³)	Radius Ratio ($\frac{r_i}{r_o}$)	Wall Thickness (μm)	E_{mb} (GPa)
0.13	0.983	0.38	1.76
0.23	0.969	0.70	3.30
0.31	0.957	0.97	4.67

Rheological and dynamic testing

An Anton Paar MCR702 dynamic mechanical analyzer (DMA) (Anton Paar, Austria) was used for the rheological testing of the specimens. Rheology measurements were performed using a parallel plate geometry with a plate diameter of 50 mm and a gap of 1 mm. The gap was selected to be significantly larger than the average particle size of 45 μm, minimizing confinement effects during testing. The experiments were conducted at shear rates from 1 to 50 s⁻¹ at 10 temperature intervals between 35-125 °C. The corresponding rotational speeds, RPM, as recorded by the rheometer are included in the dataset for completeness; however, all rheological analysis and discussion are presented in terms of shear rate in accordance with SI conventions. The data was acquired by the Anton Paar RheoCompass software. The raw data files were saved as .csv files, which are available in the repository. Figure 2 provides the experimental testing set up used, as well as the specific machine used for testing.

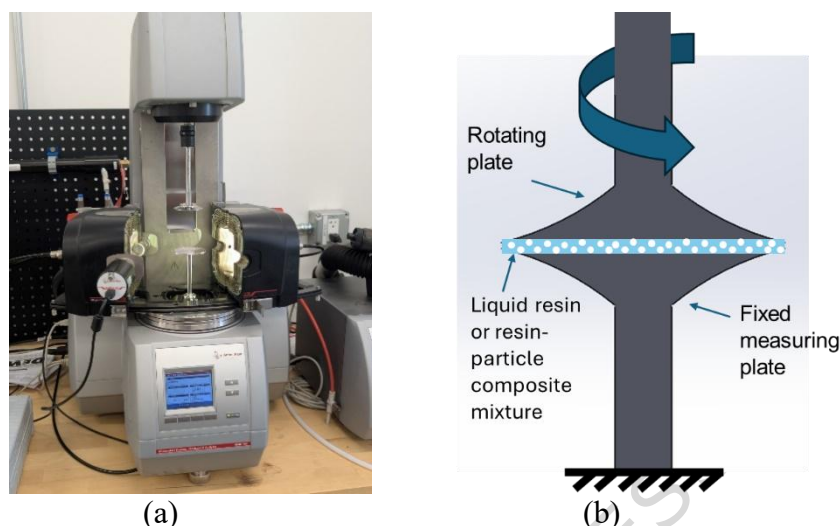


Figure 2. Rheological experimental set up (a) Anton Paar MCR702 DMA, (b) Experimental procedure with rotating plate in counter moving in clockwise direction above, stationary measuring plate below, particle filled resin is stationed in the gap between the two plates.

Temperature control during rheological testing was achieved using the integrated heating systems of the Anton Paar MCR702, which provides precise temperature regulation across the testing range 35-125 °C. Measurements were conducted under ambient laboratory conditions without the use of a solvent trap. The resin system used is a thermosetting photopolymer with low volatility over the tested temperature range, and no observable mass loss or drying effects were detected during the duration of the experiments. Additionally, the relatively short duration of each measurement minimized the likelihood of evaporation-related effects influencing the results. Prior to data acquisition, each sample was allowed to equilibrate at the target temperature to ensure thermal stability. At each measurement condition, sufficient time was allowed for the rheological response to reach a steady state before recording viscosity values. This approach minimizes the transient effects associated with temperature ramping and shear rate changes.

The resin-HGM mixtures were prepared by gradually adding HGMs into the resin at specified volume fractions, 10, 20, and 40 vol. %. Mixing was performed using gentle mechanical stirring to ensure uniform particle dispersion while minimizing breakage of suspended HGMs. The mixtures were visually inspected to confirm uniformity and were tested shortly after preparation to minimize settling effects. Prior to measurement, the sample loading and initial shear ramp were used to condition the material and promote uniform particle distribution within the measurement gap. This procedure serves as an effective preconditioning step, reducing the influence of loading history and improving measurement repeatability.

Potential wall slip effects were considered due to the use of a parallel plate geometry and particulate suspensions. The selected gap, 1 mm, is significantly larger than the particle size, reducing the likelihood of particle depletion near the boundaries. Additionally, the consistency of viscosity trends across shear rates and temperatures, and agreement between repeated measurements, indicate that wall slip effects were not significant within the tested conditions. The reproducibility of the results across multiple trials and the consistency of observed trends across

shear rates and temperatures supports the validity of the selected measurement configuration despite the constant gap selected in this work.

Microscopy and Imaging

A Skyscan 1172 from Bruker, Massachusetts, was used for the micro-computed tomography (μ CT).

Data Records

The datasets described in this study are publicly available on Zenodo. The rheological dataset is provided in a primary Zenodo deposit and contains 45 .csv files named X_XXg/cc_YY%_Resin_ViscosityZ.csv, where X is the particle density used in the formulation, Y is the volume fraction, and Z is number of the sample tested in order from 0, left blank, to 4 [22]. In addition to the composite mixtures, the dataset also includes 5 files corresponding to the neat resin, 0 vol. % HGMs, which serve as a baseline for comparison. These files follow the same naming convention, with particle density and volume fraction fields adjusted accordingly. These files were exported using Anton Paar RheoCompass. The key columns include temperature ($^{\circ}\text{C}$), viscosity (cP), shear rate (s^{-1}), and rotational speed (rpm). The temperature is separated into 10 increments from ambient temperature to 120°C . The dataset described in this work extends beyond that used in prior studies by including the full set of rheological measurements across all tested material systems and processing conditions. All raw data is provided to enable independent analysis and reuse.

In addition to the rheological data, μ CT image datasets are provided across three additional Zenodo deposits corresponding to the particle densities 0.13 , 0.23 , and 0.31 g/cm^3 [23-25]. Each μ CT deposit is organized into three .zip folders corresponding to volume fraction, 10, 20, and 40 % by vol. Each folder contains a sequence of .tif image slices representing the reconstructed 3D volume of a single specimen. These datasets are provided to support qualitative assessment of particle dispersion, agglomeration, and internal defects within the printed composites.

Data Overview

Vat photopolymerization is a photothermal process and the resin viscosity is sensitive to temperature variations [26]. Acrylate based photopolymers are known to cure poorly at lower temperatures [27]. HGMs are made of glass with substantial air cavities in their core, which makes them thermally insulating. Hence, HGMs reduce thermal conductivity of the mixture when they are dispersed in the resin, leading to localized heat retention that can cause uneven viscosity distribution during processing, especially if the HGMs are unevenly distributed [28, 29]. Viscosity is influenced by both temperature and particle loading [5]. The correlation among temperature, shear rate and viscosity is expressed by an Arrhenius-type equation [30]:

$$\xi = k_0 e^{\frac{E_a}{RT}} \dot{\gamma}^{n-1} \quad (2)$$

here $\dot{\gamma}$ is the shear rate, T is the temperature, ξ is the viscosity, R is the universal gas constant $R=8.3145 \text{ J/(K}\cdot\text{mol)}$, E_a is the activation energy, and n and k_0 are the flow behavior index, and

consistency index, respectively. For each combination of volume fraction and HGM type, five 2 ml samples were tested, totaling 50 samples. Figure 3 illustrates the temperature and viscosity profile of the neat resin from the median sample of the dataset. At shear rates below approximately 10 s^{-1} , the measured torque approaches the lower operational limit of the rheometer, leading to increased uncertainty and variability in the recorded viscosity values. This results in the apparent spread observed in Figure 3. As the shear rate increases, the signal-to-noise ratio improves and the viscosity response stabilizes, confirming the neat resin exhibits near-Newtonian behavior in the reliable measurement regime. This behavior is consistent with expected limitations in rotational rheometric at low shear rates, where the ratio of signal-to-noise decreases due to low torque values. The linear scales used in Figure 3 are to preserve visibility of low shear rate behavior and measurement variability, which are important for interpreting the full dataset across the testing range.

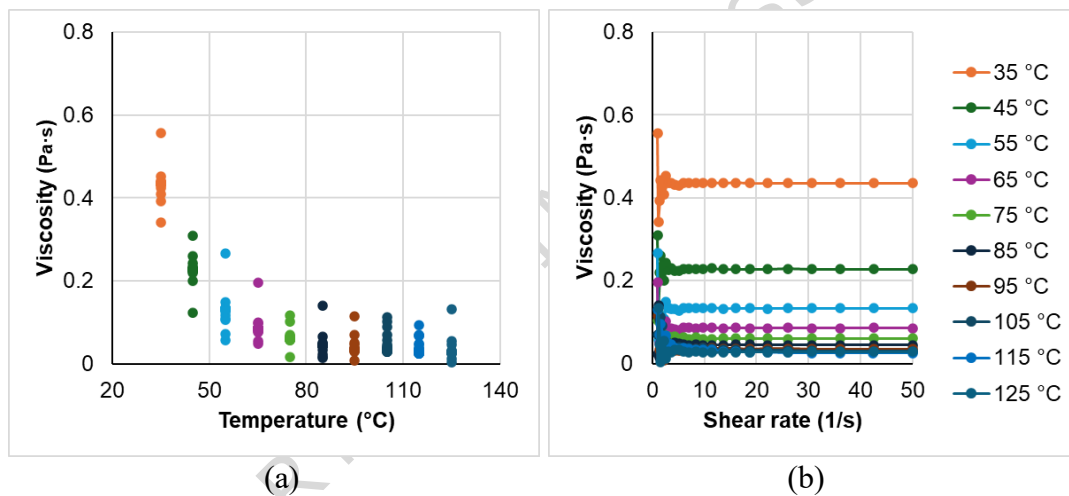
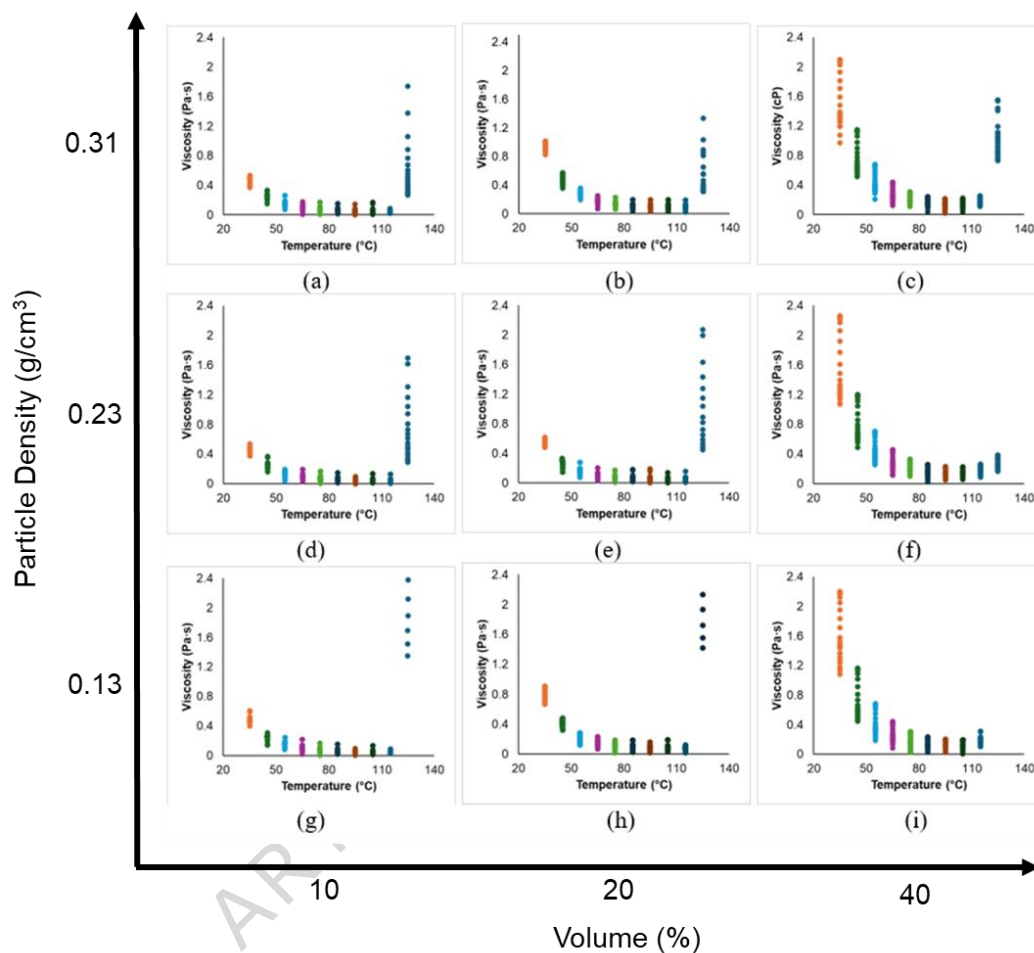


Figure 3. Graphs depicting (a) the viscosity versus temperature for neat resin, and (b) the viscosity versus shear rate at various temperatures. Increased variability at low shear rates is due to measurement instability associated with low torque sensitivity limits of the rheometer. At higher shear rates, the measurements stabilize, showing near-Newtonian behavior in the neat resin.

Comparison of the three HGM grades revealed consistent overall trends that the viscosity decreased as the temperature increased but with notable differences in magnitude and processing



range.

Figure 4 shows that viscosity increased with HGM content across all particle types. At elevated temperatures, above 120 °C, an increase in viscosity is observed across all samples, although the onset and magnitude vary with volume fraction [31]. No evidence of material loss, edge fracture, or sample expulsion from the measurement geometry was observed during testing. The viscosity of neat resin decreased from 0.45 to 0.03 Pa·s with an increase in temperature from room temperature to 125 °C as seen in Figure 3(a). This resin is designed to be used at room temperature. The neat resin exhibited a viscosity of approximately 0.40 Pa·s near room temperature, which corresponds to the manufacturer-recommended printing condition for the neat resin. This value was therefore used as a benchmark for identifying temperature ranges at which the resin-HGM mixtures exhibit comparable flow behavior. The purpose of this benchmark is to support relative process mapping across formulations, rather than to imply that viscosity alone fully defines printability. The addition of HGMs increases the viscosity of the resin relative to the neat system, requiring adjustment of temperature to achieve viscosity values comparable to the reference processing condition. Mixtures containing 10, 20, and 40 vol.% HGMs exhibited stable viscosities of 0.40 Pa·s at ~45–65 °C, 45–55 °C, and 65–75 °C temperature ranges, respectively, though the precise range varied by density grade.

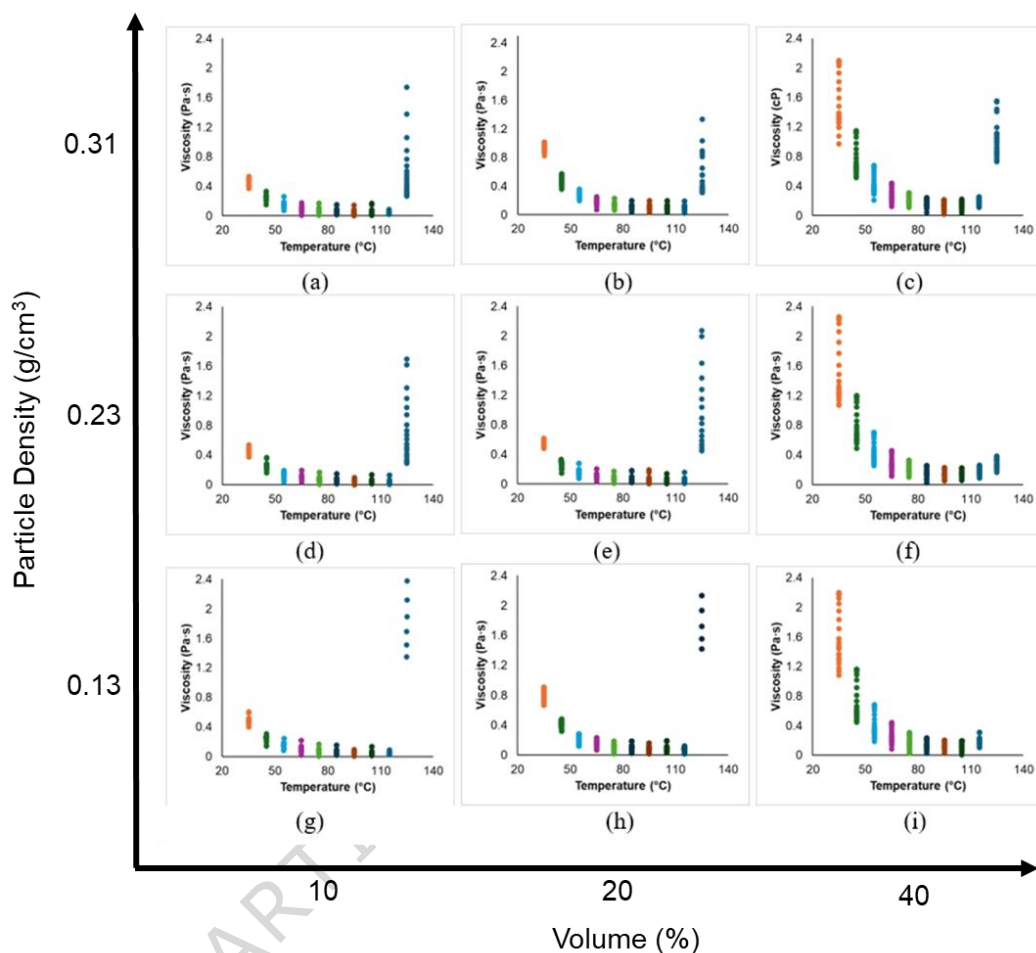


Figure 4. Viscosity–temperature profiles for resin loaded with HGMs of (a–c) 0.31 g/cm³, (d–f) 0.23 g/cm³, and (g–i) 0.13 g/cm³ at 10, 20, and 40 vol.%. The vertical spread of data is related to various shear rates.

Shear rate dependence of viscosity is clearly illustrated in Figure 5. For neat resin, Figure 3(b), viscosity remained nearly constant across the tested shear rates. In contrast, resin–HGM mixtures exhibited shear-thickening behavior up to approximately 20 s⁻¹, followed by a plateau in viscosity at higher shear rates. Comparison of the three particle types shows that suspensions with thicker-walled microspheres reach this plateau at higher shear rates than those with thinner-walled microspheres, particularly at elevated temperatures, indicating stronger particle–resin interactions for thicker-walled particles.

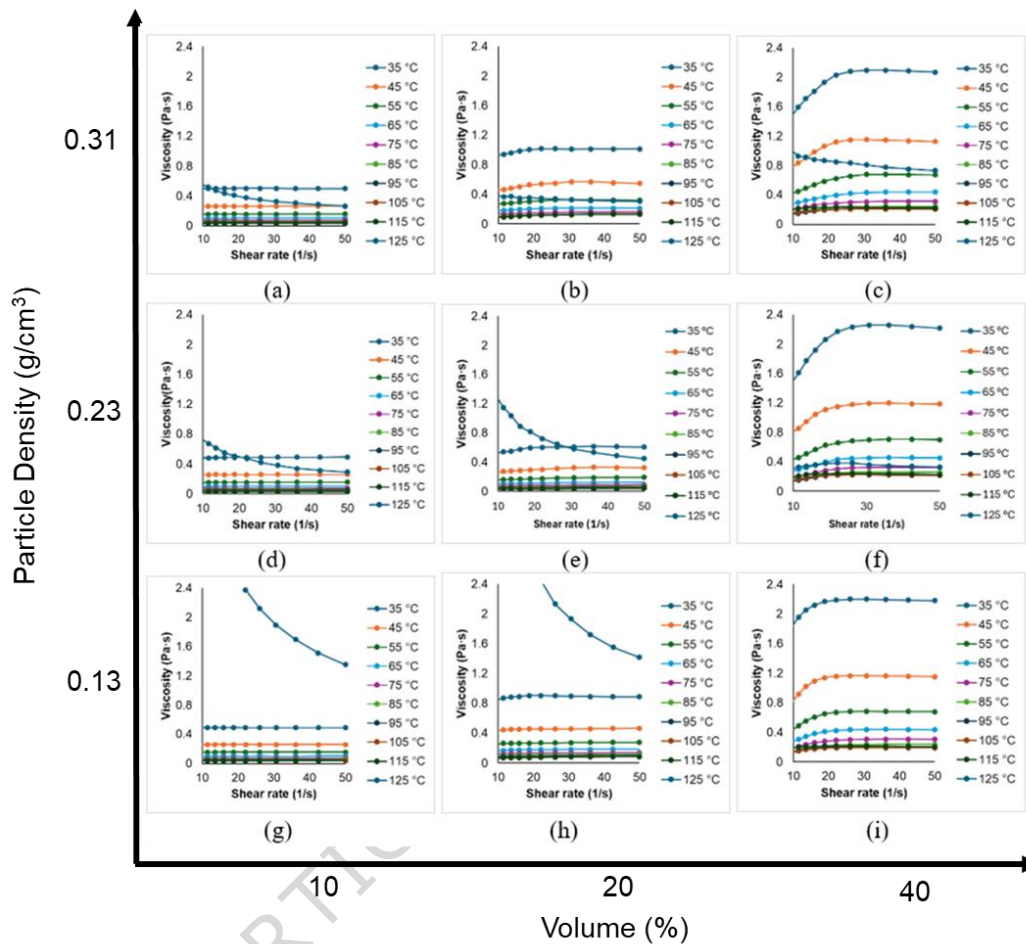


Figure 5. Viscosity–shear rate behavior at multiple temperatures for resin loaded with HGMs of (a–c) 0.31 g/cm³, (d–f) 0.23 g/cm³, and (g–i) 0.13 g/cm³ at 10, 20, and 40 vol.%.

Usage Notes

In the present case, the ideal processing temperature can be identified for each mixture. Hence, the shear rate is of interest only at that temperature. For all syntactic foams, a shear rate of 20 s⁻¹ was identified as optimal, corresponding to ~2.62 rad/s for the chosen parallel-plate setup. The RPM as recorded by the Anton Paar for the respective shear rates is also captured in the dataset. The dataset provides a mapping of viscosity as a function of temperature and shear rate for different HGM loadings, enabling identification of processing conditions suitable for specific applications. Rather than prescribing a single set of optimal parameters, the data are intended to support user-defined process selection based on application and equipment constraints. The volume fractions of HGMs are selected based on the desired final material properties, including density and mechanical performance, and are not adjusted to achieve printability. Instead, the interpolation approach is used to identify temperature and shear rate conditions that enable processing of a fixed formulation. This ensures that the target material properties are maintained while achieving appropriate rheological behavior for additive manufacturing.

Technical validation

Rheological measurements were conducted across a controlled range of temperatures, 35–125 °C and shear rates 1–50 s⁻¹ using standardized test protocols to ensure repeatability and consistency. For each material system measurements were performed under identical conditions, and datasets were evaluated for continuity and physical consistency by examining expected monotonic trends in viscosity with respect to temperature and shear rate, as well as agreement across repeated trials for each material composition. Outliers and anomalous data points were evaluated based on deviations from expected trends and consistency with adjacent measurement points. Data points were excluded only when such deviations were attributable to identifiable measurement artifacts, such as instability at low shear rates due to torque sensitivity limits of the rheometer.

To further assess data reliability, viscosity values were compared across overlapping temperature measurements and shear rate conditions to confirm consistency between adjacent measurement regimes. Interpolation across datasets was used as an internal check to verify smooth transitions and absence of discontinuities, indicating stable rheological behavior during testing. Additional non-destructive testing methods such as μ CT have been used to qualitatively assess particle dispersion and detect potential defects within materials [32, 33].

μ CT scans were performed using a Bruker Skyscan 1172. Specimens were printed to 10 mm diameter by 5 mm tall cylinders suitable for the scanning chamber and mounted on a brass stage with a light adhesive to minimize movement during acquisition. Scans were conducted at a source voltage of 10 kV, and current of 10 μ A, with a camera pixel size of 9 μ m. A rotation step of 0.35° was used over a 360° scan range.

Reconstruction was carried out using NRecon software with beam hardening correction and ring artifact reduction applied to improve image quality. The reconstruction volumes were analyzed using grayscale thresholding to distinguish hollow glass microspheres from the surrounding resin matrix, enabling qualitative assessment of particle dispersion and identification of potential defects.

μ CT was used to evaluate the quality of additively manufactured specimens corresponding to the rheological datasets. Scans were performed on representative specimens across the full range of particle wall thicknesses and volume fractions. The reconstructed volumes showed uniform particle dispersion and no significant agglomeration or sedimentation, supporting the assumption of homogeneous mixtures during rheological testing. No internal voids or defects were observed in neat resin specimens through optical microscopy, further confirming material consistency. Representative slices from multiple regions of each specimen were examined to assess spatial uniformity of particle distribution. No evidence of significant agglomeration, sedimentation gradients, or particle-depleted regions was observed within the resolution limits of the scans. Depicted in Figure 6 and Figure 7 are image slices of syntactic foams of the two extremes for volume fractions. These images show uniform particle distribution in the specimens.

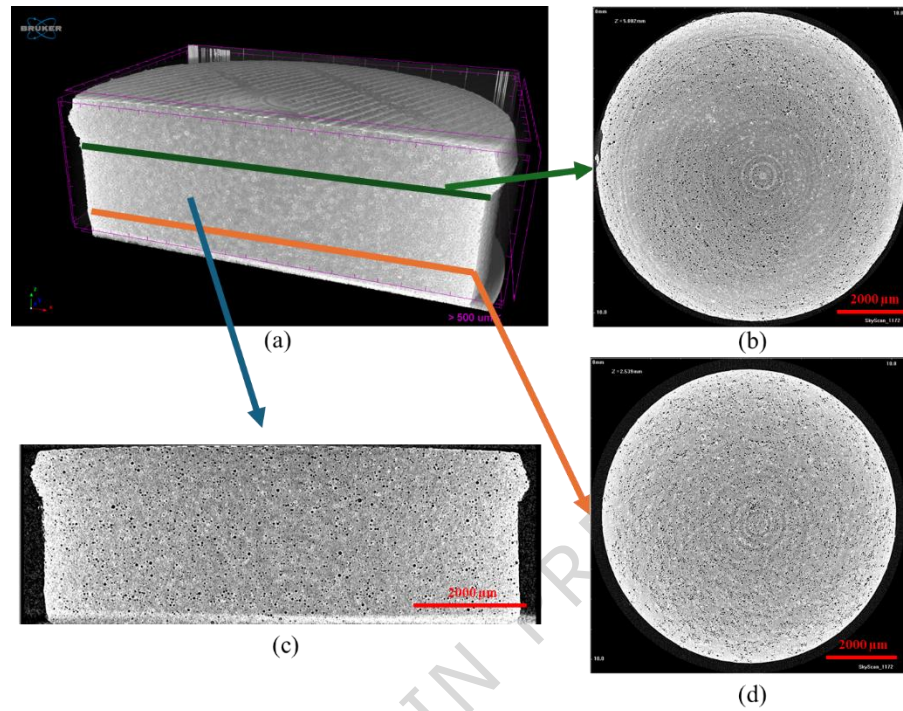


Figure 6. μ CT scan image slices of a 0.31 g/cm³ 10% by vol specimen: (a) cross-section, (b) near the top of the specimen, (c) coronal view, and (d) near the bottom of the specimen.

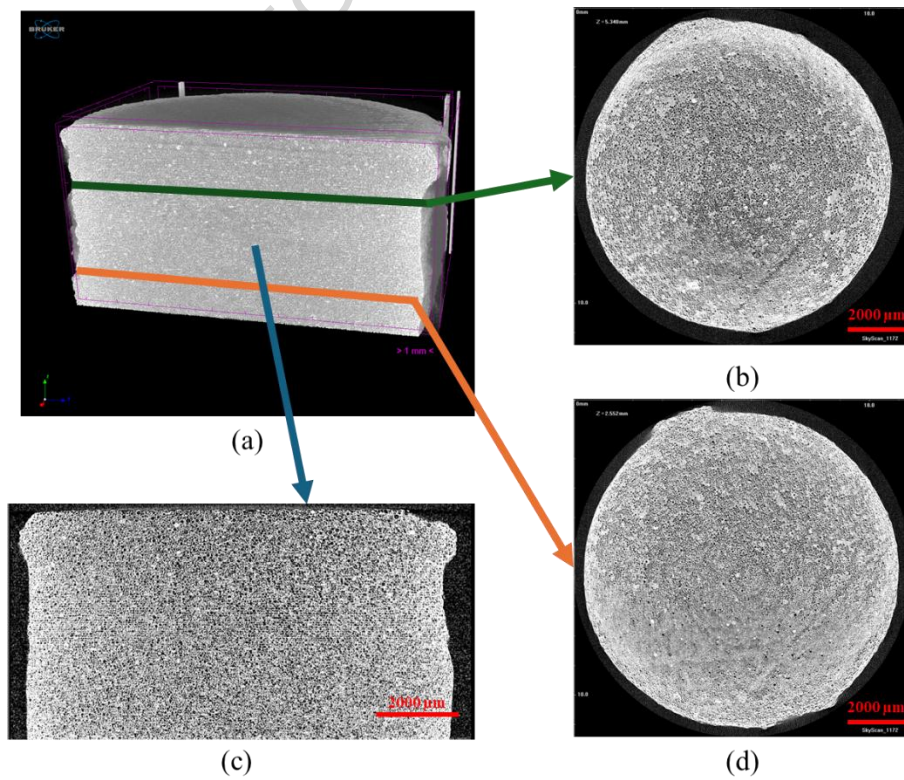


Figure 7. μ CT scan image slices of a 0.23 g/cm³ 40% by vol specimen: (a) cross-section, (b) near the top of the specimen, (c) coronal view, and (d) near the bottom of the specimen.

Additionally, the experimental methodology and resulting datasets are consistent with prior validation studies on similar material systems, where rheology-guided processing conditions produced reproducible mechanical performance [18].

Code Availability

The authors declare that no code was used in the creation of this data.

Data Availability

The data supporting this study are publicly available on Zenodo. The rheological data is available under DOI <https://doi.org/10.5281/zenodo.19699588>. The associated μ CT datasets are under DOIs <https://doi.org/10.5281/zenodo.19699628>, <https://doi.org/10.5281/zenodo.19699671>, <https://doi.org/10.5281/zenodo.19700670>, corresponding to particle densities 0.13, 0.23, and 0.31 g/cm³, respectively. All datasets are fully accessible without restriction and are distributed under a Creative Commons Attribution 4.0 International license.

Authors Contributions

According to CrediT contributorship taxonomy, Caleb Beckwith performed roles of conceptualization, formal analysis, investigation, resources, writing - original draft, writing - reviewing and editing, and visualization. Nikhil Gupta performed the roles of conceptualization, formal analysis, investigation, writing – reviewing and editing, visualization, supervision, project administration, and funding acquisition.

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Funding Declaration

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Declaration of Competing Interests

The authors declare no competing interests or conflict of interests.

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